Version Control System — Assignment

Git Conflict Simulation (Two Developers, One File)

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Subject	Version Control System
Assignment	Git Merge Conflict Simulation

Task Objective

Two Developers, One File Student simulates two developers modifying the same file, identifies the conflict, resolves it, and successfully merges both contributions.

Step-wise Commands

Step 1: Create Repository and File

Initialize a new Git repository and create the demo file with its original content.

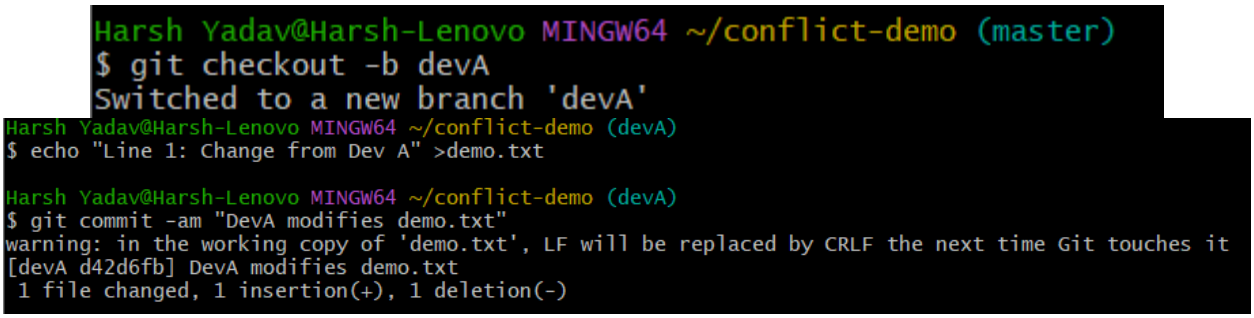
```
git init conflict-demo
cd conflict-demo
echo "Line 1: Original content" > demo.txt
git add demo.txt
git commit -m "Initial commit with demo.txt"
```



Step 2: Create Developer A Branch

Create a new branch for Developer A and make a change to the file.

```
git checkout -b devA
echo "Line 1: Change from Developer A" > demo.txt
git commit -am "DevA modifies demo.txt"
```



Step 3: Create Developer B Branch

Switch back to master, then create a separate branch for Developer B with a conflicting change.

```
git checkout master
git checkout -b devB
echo "Line 1: Change from Developer B" > demo.txt
git commit -am "DevB modifies demo.txt"
```



Step 4: Merge DevA into Master

Switch to the master branch and merge Developer A's changes in.

```
git checkout master
git merge devA
```

```
Harsh Yadav@Harsh-Lenovo MINGW64 ~/conflict-demo (master)
$ git merge devA
Updating 382825c..d42d6fb
Fast-forward
 demo.txt | 2 +-
 1 file changed, 1 insertion(+), 1 deletion(-)
```

Step 5: Merge DevB into Master (Conflict Trigger)

Attempt to merge Developer B's branch into master. Since both branches modified the same line, this produces a merge conflict in demo.txt.

```
git merge devB
```

```
Harsh Yadav@Harsh-Lenovo MINGW64 ~/conflict-demo (master)
$ git merge devB
Auto-merging demo.txt
CONFLICT (content): Merge conflict in demo.txt
Automatic merge failed; fix conflicts and then commit the result.
```

Step 6: Resolve Conflict

Open demo.txt in a text editor. Git marks the conflicting sections as shown below:

```
<<<<<<< HEAD
Line 1: Change from Developer A
=======
Line 1: Change from Developer B
>>>>>>> devB
```

Edit the file to resolve the conflict, for example:

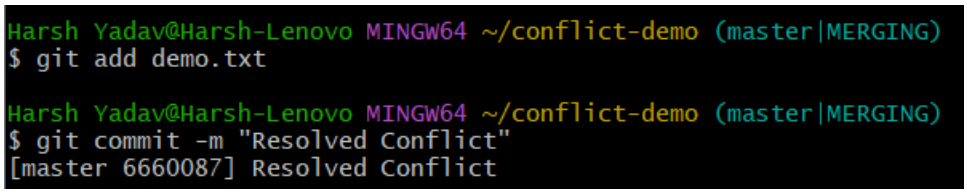
```
Line 1: Change from Developer A and Developer B
```

```
Harsh Yadav@Harsh-Lenovo MINGW64 ~/conflict-demo (master|MERGING)
$ notepad demo.txt
```

Step 7: Finalize Merge

Stage the resolved file and commit to complete the merge.

```
git add demo.txt
git commit -m "Resolved conflict"
```



End Result

- Simulated two developers editing the same file.
- Triggered a merge conflict.
- Resolved it manually.
- Committed the resolution.